1 Basic Point-Set Topology

Throughout these notes we fix a topological space X.

Definition 1. A/an (open) **neighbourhood base at** $x \in X$ is a collection of (open) sets $N_x \subset \mathcal{T}$ such that:

- $x \in V$ for all $V \in N_x$, and
- If U open, $x \in U$ (we sometimes write this as *if U is in the neighbourhood of x*), then there exists $V \in N_x$ such that $V \subseteq U$.

A **base** of the topology $\mathcal{T}$ is the set $\mathcal{B} := \bigcup_{x \in X} N_x$. It follows that, given a base $\mathcal{B}$, the set $M_x := \{N : N \in \mathcal{B}, x \in N\}$ is a neighbourhood base at x.

Theorem 1. *The following are equivalent:*

1. *Every non-empty open set is a union of sets in $\mathcal{B}$.*
2. *$\mathcal{B}$ is a base for a topology.*
3. - *For all $x \in X$, there is $A \in \mathcal{B}$ such that $x \in A$*
   - *If $U, V \in \mathcal{B}$ and $x \in U \cap V$, then there exists $W \in \mathcal{B}$ such that $x \in W$ and $W \subseteq U \cap V$.*

Definition 2. A basis for the **product topology** is $\{U \times V : U \in X, V \in Y \text{ open}\}$.

Theorem 2. *$f : Z \to X \times Y$ continuous if and only if $p_1 \circ f : Z \to X$, $p_2 \circ f : Z \to Y$ are continuous.*

Definition 3. The **quotient topology** $X/\sim$ is where considering the canonical projection $q : X \to X/\sim$, we have $\widetilde{U} \subseteq X/\sim$ is open if and only if $q^{-1}(\widetilde{U})$ is open in X.

Theorem 3 (Descending Quotients)**.** *If $f : X \to Y$ is a continuous function such that $x_1 \sim x_2 \Rightarrow F(x_1) = F(x_2)$, then f is immediately continuous in the quotient topology.*

Definition 4. A space is **connected** if it can not be written as the union of two disjoint open sets, it is **path connected** if there is a **path** (a continuous function from $[0, 1]$ to the space) with starting point and end point to be any two points in the space.

Definition 5. X is **locally path-connected** if for all $x \in X$, and any open neighbourhood U of x, there is V open and path-connected such that $x \in V \subseteq U$.

Theorem 4. *Connected and locally path connected implies path-connected.*

A proof can be done by thinking about the path component that contains x is open. why is it closed?

Theorem 5. *A space is connected if and only if the only closed and open sets are the empty set and itself.*

Theorem 6. *A closed subspace of a compact/Hausdroff space is compact/Hausdroff, a surjective continuous map sends compact space to compact space. A compact subspace of a Hasudroff space is closed.*

Theorem 7. *If $f : X \to Y$ with X compact, Y Hausdroff, f continuous and bijective, then f is a homeomorphism .*

2 CW Complexes and Homotopies

Let $\overline{\mathbb{D}}^n := \{v \in \mathbb{R}^n : \|v\| \leq 1\}$, we will call this a n-cell. Note that it is a n-dimensional manifold with boundary and write $\overline{\mathbb{D}}^n = \mathbb{S}^{n-1} \cup \mathbb{D}^n$

Definition 6. Given a topological space X, a continuous function $\phi : \mathbb{S}^{n-1} \to X$ the space of **attaching a n-cell by ϕ** denoted by $X \cup_\phi \overline{\mathbb{D}}^n$, is the space $X \coprod \overline{D}^n/\sim$, equivalence relation given by $x \sim y$ if and only if $x = y$ or $\phi(x) = y$ or $\phi(y) = x$. That is, it only identifies the boundary with the image.

Definition 7. A **CW complex** is a space X built by the following inductive process:

- Start at $X^{-1} = \emptyset$, and attach a collection of 0-cells. The result is a discrete space (just a bunch of points) called X^0 (the **0-skeleton** of X).

- You add collections of 1-cells $(\overline{\mathbb{D}^1})_i$ (possibly infinitely many) by specifying attaching maps $\phi_i^1 : (\mathbb{S}^0)_i \to X^0$. The result is called the **1-skeleton** X^1.
- You add collections of 2-cells $(\overline{\mathbb{D}^2})_j$ (possibly infinitely many) by specifying attaching maps $\phi_j^2 : (\mathbb{S}^1)_j \to X^1$. The result is called the **2-skeleton** X^2.

Continue and declare $X = \bigcup_{n\geq 0} X^n$ and equip it with the weak topology, (in which a subset $U \subset X$ is open if and only if $U \cap X^n$ is open for all $n \geq 0$ - this part we won't care too much). If the process above stops after n iterations for some n, it is called a n**-dimensional CW complex**.

So note that to specify a 2D CW complex we would need information on the collection of 0 cells, ϕ_i^1, and ϕ_j^2s.

Definition 8. Let $A \subseteq X$ A **deformation retraction to** A is a map $F : X \times [0,1] \to X$ such that

- F is continuous.
- $F(x,0) = x$ for all $x \in X$
- $F(x,1) \in A$ for all $x \in X$
- $F(a,t) = a$ for all $a \in A, t \in [0,1]$.

Definition 9. Let $f_0, f_1 : X \to Y$, a **homotopy between** f_0 **and** f_1 is $F : X \times [0,1] \to Y$ such that:

- F is continuous,
- $F(x,0) = f_0(x)$ for all $x \in X$,
- $F(x,1) = f_1(x)$.

We write $f_0 \simeq f_1$ if such F exists.

Equivalently, we can think about homotopy as a family of functions $\{f_t\}_{t\in[0,1]}$, keep in mind that $F(x,t) = f_t(x)$.

Definition 10. A **path homotopy** between two paths, $\gamma_0, \gamma_1 : [0,1] \to X$, is only defined when $\gamma_0(0) = \gamma_1(0), \gamma_0(1) = \gamma_1(1)$. It is a homotopy $F : [0,1] \times [0,1]$ between γ_0, γ_1, with the extra requirement that $F(0,t) = F(0,0), F(1,t) = F(1,0)$ for all $t \in [0,1]$. (Intuitively, it is a continuous change such that the start and end points are fixed)

Definition 11. $f : X \to Y$ is a **homotopy equivalence** if there is a **homotopy inverse**: that is, a function g such that $g \circ f \simeq id_X, f \circ g \simeq id_Y$. We write $X \simeq Y$ if such f, g exists.

Theorem 8. *$\simeq$, in both senses (functions and spaces), is an equivalence relation.*

A special case to understand is retraction.

Definition 12. A **retraction to** A is a continuous map $F : X \to A$ such that $F(a) \in A$ for any $a \in A$.

Theorem 9. *If $F : X \times [0,1] \to X$ is a deformation retraction to A, then*

- *$r(x) = F(x,1)$ is a retraction to A,*
- *F is a homotopy equivalence with the pair (i, r), where i is the inclusion from A to X.*

All this is saying a deformation retraction yields a homotopy.

Another special case is **graphs** (1-dimensional CW complexes), and they have a special property:

Definition 13. The **homotopy extension property for a pair** (X, A), where $A \subseteq X$ is such that for any continuous function $F : X \to Y$, any homotopy $h : A \times [0,1] \to A$ with $h(a,0) = F(a)$ has a homotopy $H : X \times [0,1] \to X$ with $H(x,0) = F(x)$ and $H|_{A\times[0,1]} = h$.

Intuitively, any homotopy of X is determined by the homotopy on A.

Theorem 10. *Any graph X has the HEP if A is a choice of a complex that visits through any vertices once.*

Theorem 11. *If (X, A) has the HEP, then $X \simeq X/\sim$, where $\sim$ identifies everything in A into one point.*

3 Groups and Fundamental Things

Definition 14. Let $x_0 \in X$. We define the **(first) fundamental group** $\pi_1(X, x_0) = \{\text{Paths Starting and ending at } x_0\}/$ path-homotopies. The group operation is concatenation of paths.

Note that any based map $f : (X, x_0) \to (Y, y_0)$ **induces** a map of fundamental groups

$$f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0), \qquad [\gamma] \mapsto [f \circ \gamma]$$

, the main point of the rest of this note is to study what is the image and kernel of these homomorphisms.

Definition 15. Given a path h in X starting in x_1, ends at x_-, one can produce a **change-of-basepoint isomorphism** $\beta_h \pi_1(X, x_0) \to \pi_1(X, x_1)$ by $[\gamma] \to [h \bullet \gamma \bullet h^{-1}]$ (Note: concatenation is read from left to right)

Theorem 12 (Seifert-Van Kampen)**.** *Let $U, V \subseteq X$ open and $x_0 \in U \cap V, U \cup V = X$, then*

$$\pi_1(X, x_0) = \pi_1(U, x_0) *_{U \cap V} \pi_1(V, x_0).$$

Theorem 13. *If $A \subseteq X$ such that $x_0 \in A$, and $r : X \to A$ is a retraction, then consider the inclusion $i : A \to R$, we have $i_* : \pi_1(A, x_0) \to \pi_1(X, x_0)$ is injective.*

The free group on a set S can of course be defined by words, but the following *universal property* is useful:

Definition 16. Given a set S, the **free group on** S, denoted by $F(S)$, is a *group* that has the two properties:

- $S \subseteq F(S)$, that is, there is an injective set-function $i : S \to F(S)$, and
- For any group G and any set-function $f : S \to G$, there is a unique group homomorphism $f^! : F(S) \to G$ such that $f = f^! \circ i$.

One of course need to check this defines a group up to isomorphism, but after that, we can immediately use Theorem 11 to conclude the wedge of circles $B(a_1, a_2, ...a_n)$ has fundamental group $F(\{1, ...n\})$ (more commonly called F_n).

For the rest of the notes, we turn our focus to the fundamental group of 1-dimensional and 2-dimensional CW complexes. We assume the complexes are connected, moreover, with the extra property that CW complexes are always locally path-connected, it is always path-connected.

Definition 17. A **locally injective** (continuous) map is $f : X \to Y$ such that for any $x \in X$ there is O_x open such that $x \in O_x$ and $f|_{O_x}$ is injective. Similarly, one can define **locally surjective**.

Definition 18. Let X, Y be graphs. A **map of graphs** is a continuous map $f : X \to Y$ such that vertices are mapped to vertices, and edges are mapped to edges *homeomorphically*.

That is, to specify a map of graphs, you label the vertices and edges of Y first, then put labels on X such that the orientation of labels is consistent with what is labeled already in Y.

Locally injective (resp. surjective) map of graphs means counting the number of edges going in and out for *each* vertices and making sure they have at least one (resp. all) of the edges present. Here are the three key theorems, set X to be a connected graph:

Theorem 14. *There is a set T such that X is homotopic equivalent to $B(T)$.*

Theorem 15. *For any $H \subseteq F(S)$, there is a connected graph X such that there is a locally injective map of graphs $f : X \to B(S)$ such that $im(f_*) = H$*

Theorem 16. *If $f : X \to B(S)$ is locally injective. then f_* is injective. That is, $\pi_1(X) \cong im(f_*)$.*

Combining all 3 shows every subgroup of a free group is free.